

APURBA AFIAT

🏠 Mirpur DOHS, Dhaka, Bangladesh 📞 01868112233 ✉ apurba.afiati@gmail.com
🌐 [apurba-afiati](#) 👤 [ApurbaAfiati](#) 🌐 [apurbaafiati.github.io](#)

PROFILE STATEMENT

Computer Science graduate with experience in machine learning, backend development, and data centric applications. Skilled in Python, AI frameworks, and modern web technologies, with practical exposure to building scalable systems and AI enabled solutions. Seeking to contribute technical expertise and problem solving skills in a fast paced corporate environment.

EXPERIENCE

Life Plus Bangladesh Ltd.

April '26 - Present

Software Quality Assurance Engineer

Dhaka, Bangladesh

- Improved software quality by designing and executing detailed test plans, test cases, and scenarios, resulting in more reliable and user-friendly applications.
- Reduced production defects by performing manual and automated testing, identifying bugs early, and collaborating with developers for timely resolution.
- Enhanced system performance and compliance by validating functionality, usability, and data integrity, ensuring adherence to QA standards and healthcare requirements.

Avian™: Bangladesh's Trusted Tech Brand

Feb '26 - Mar '26

Software Quality Assurance Engineer

Dhaka, Bangladesh

- Executed manual testing for web-based and internal applications, ensuring functionality and usability, resulting in enhanced product quality.
- Identified and reported bugs with clear steps to reproduce and relevant details, leading to faster issue resolution.
- Assisted in regression and sanity testing, ensuring the stability and readiness of the product before deployment.

Brainiaxs-IT

Mar '25 – May '25

Wordpress Developer

Dhaka, Bangladesh

- Managed end-to-end WordPress website development, including custom themes and plugins, for diverse client requirements.
- Delivered high-performance websites that improved user engagement, with a **30% increase in average session duration**.
- Utilized WordPress, PHP, JavaScript, HTML, CSS, and MySQL to ensure responsive design and seamless user experience.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, Java, PHP, C++
- **Testing and Quality Assurance:** Postman, Performance and Stress testing, Jira, Manual Testing(Test case design, test scenarios, exploratory testing), Unit Testing(Jest), Selenium, JMeter
- **AI & ML Tools:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Data Analytics:** Tableau, Power BI
- **Databases:** MySQL, MongoDB
- **Backend Development :** Node.js, Express, Flask, Django, REST APIs
- **Web Development:** HTML, CSS, Javascript, React, Bootstrap
- **Other Expertise:** Git, Data Scraping, Canva, Professional Video editing, Microsoft tools(excel, word, pptx)

EDUCATION

BRAC University

Jul '21 – Nov '25

Bachelor of Science in Computer Science

Dhaka, Bangladesh

Military Collegiate School Khulna

Mar '13 – May '20

Science background

Khulna, Bangladesh

PROJECTS

Bilingual WASH-QA: Enhancing Health Data Access with Multilingual AI | *Python, FAISS, RAG, LLM* Jul '24

- Built a bilingual Question Answering system supporting Bangla and English queries for health and sanitation data under the WHO/UNICEF JMP framework, using Retrieval-Augmented Generation (RAG) and multilingual sentence embeddings.
- Preprocessed and structured multiple WASH Excel datasets into a QA-friendly format, integrated FAISS for similarity-based retrieval, and used SentenceTransformer (distiluse-base-multilingual-cased-v1) for vector embeddings.
- Enabled real-time translation of Bangla queries using Deep Translator with Google Translate API and addressed Unicode rendering challenges with UTF-8 encoded output.
- Proposed improvements including custom translation models, fine-tuned embeddings, and adding voice/dialect support to enhance accessibility.

Policy Document AI Chatbot: Advanced RAG for Financial Policy Analysis | *Python, ChromaDB, LLM* Aug '25

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) system to query and analyze complex government financial policy PDFs, enabling precise, context-aware Q&A with accurate page citations.
- Built a FAISS-based vector database using ChromaDB and the BGE-small-en-v1.5 sentence transformer model for efficient and accurate semantic similarity search.
- Implemented a conversational memory system to maintain context across user turns, using query augmentation and keyword tracking to significantly improve multi-turn dialogue coherence.

Ecommerce Web Application | *MongoDB, Express.js, React, Node.js, JWT, REST API* Jan '26

- Built a full-featured ecommerce platform supporting user authentication, product management, cart, order processing, and admin dashboard.
- Integrated secure JWT-based authentication, product search, reviews, and payment gateway support.
- Implemented responsive UI with React and Redux, and RESTful backend with Node.js/Express and MongoDB.
- Deployed with environment-based configuration, robust error handling, and currency localization for Bangladeshi Taka.

Parking Management System | *Python, HTML, CSS, PHP, MYSQL* Sep '23

- Developed a parking management system to streamline parking allocation and payment processes. The system informs users about available parking slots in real time and facilitates seamless digital payments. Designed to enhance efficiency and reduce congestion, providing a user-friendly solution similar to the university's parking system.

Detection of Monkeypox using Deep Learning | *Python, Deep Learning, CNN, TensorFlow* Feb '25

- Engineered a deep learning model using Convolutional Neural Networks to classify skin lesions and detect Monkeypox from image data. Achieved promising accuracy by preprocessing datasets and fine-tuning model hyperparameters.

ACHIEVEMENTS

AgentX BD Competition 2025 October '25

Top 9 Finalist (out of 2000 participants)

Netcom Learning, BRAC University & Microsoft

- Developed an AI Agent named **Aparajita**, designed to support women in Bangladesh through empathetic, big-sister-style conversations.
- Trained the agent on Bangladesh's legal rights, emergency contacts, NGO information, hospitals, and coping strategies for stress and loneliness.
- Ranked **9th among 2000 participants** for innovation and social impact.

LEADERSHIP / EXTRACURRICULAR

BRAC University Community Service Club Nov '23 – Dec '24

Vice President

BRAC University

****Led 300 members to successfully organize community service initiatives, including blood donation drives, Ramadan ration distributions, book drives, saline and drinking water distribution during heatwave, organising awareness seminar, vaccination to stray animals which impacted numerous beneficiaries.****

LEAD Academy Mar '24 - Dec '24

Campus Ambassador

BRAC University

Provided guidance and facilitated access to valuable resources and courses for students who are interested in exploring specific subjects to help them understand the educational needs and career paths.